

Survey / Research Paper
Project Title: Predictive Healthcare Analytics

Prepared By
Khyati Vinodbhai Jasani
Post Graduate Diploma in Data Science Student
(Year : 2025-26)
(Enrolment No.: 251370680004)
Internal Guide: Prof. Mahesh Panchal
Research Guide: Prof. Dr. Seema Joshi

Submitted to :



GUJARAT TECHNOLOGICAL UNIVERSITY
SCHOOL OF ENGINEERING AND TECHNOLOGY
CHANDKHEDA, AHMEDABAD

Abstract

Predictive Healthcare Analytics is an emerging field that combines healthcare data and machine learning techniques to improve disease prediction and patient care. With the rapid growth of medical datasets, predictive models can help identify diseases at early stages, reduce healthcare costs, and support evidence-based decision-making. This paper presents a brief survey of existing healthcare prediction methods and proposes a machine learning-based predictive healthcare system using laboratory blood parameters. The study compares various machine learning algorithms such as Naive Bayes, J48 Decision Tree, and Random Forest for disease prediction. Among these models, Random Forest demonstrated the highest prediction accuracy.

1. Introduction

Healthcare organizations generate enormous amounts of patient and laboratory data every day. Traditional methods of disease diagnosis are often time-consuming and depend heavily on manual analysis by medical experts. Predictive Healthcare Analytics uses machine learning algorithms to analyze healthcare data and identify hidden disease patterns automatically.

Machine learning techniques can assist doctors in:

- Early disease detection
- Predictive risk assessment
- Personalized treatment planning
- Reducing healthcare costs
- Improving patient monitoring systems

This project focuses on predicting health risks using blood parameter datasets collected from healthcare records. By applying machine learning techniques, the system can classify patients based on disease probability and support faster clinical decisions.

2. Literature Survey / Existing Methods

Several researchers have applied machine learning algorithms in healthcare analytics for disease prediction and medical diagnosis.

2.1 Naive Bayes Algorithm

Naive Bayes is a probabilistic classification algorithm based on Bayes' theorem. It assumes independence among input features and is widely used for medical diagnosis because of its simplicity and fast execution.

Advantages

- Fast and efficient
- Works well with large datasets
- Easy to implement

Limitations

- Assumes feature independence
 - Accuracy may decrease with complex datasets
-

2.2 Decision Tree (J48)

J48 is a decision-tree classification algorithm used to create predictive models by splitting datasets into branches based on attribute values.

Advantages

- Easy to understand and visualize
- Good classification performance
- Handles missing values effectively

Limitations

- Can overfit the dataset
 - Performance depends on data quality
-

2.3 Random Forest Algorithm

Random Forest is an ensemble machine learning method that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Advantages

- High prediction accuracy
- Handles large datasets effectively
- Reduces overfitting problems

Limitations

- Higher computational complexity
- Less interpretable compared to single decision trees

3. Problem Statement

Modern healthcare systems face challenges such as:

- Delayed disease detection
- Large volumes of laboratory data
- Increasing healthcare costs
- Lack of predictive analysis systems

Manual analysis of patient records may lead to slower diagnosis and reduced efficiency. Therefore, there is a need for intelligent predictive healthcare systems capable of analyzing patient laboratory parameters automatically.

4. Proposed System

The proposed system uses machine learning algorithms to predict diseases using blood parameter datasets.

Workflow of Proposed System

Step 1: Data Collection

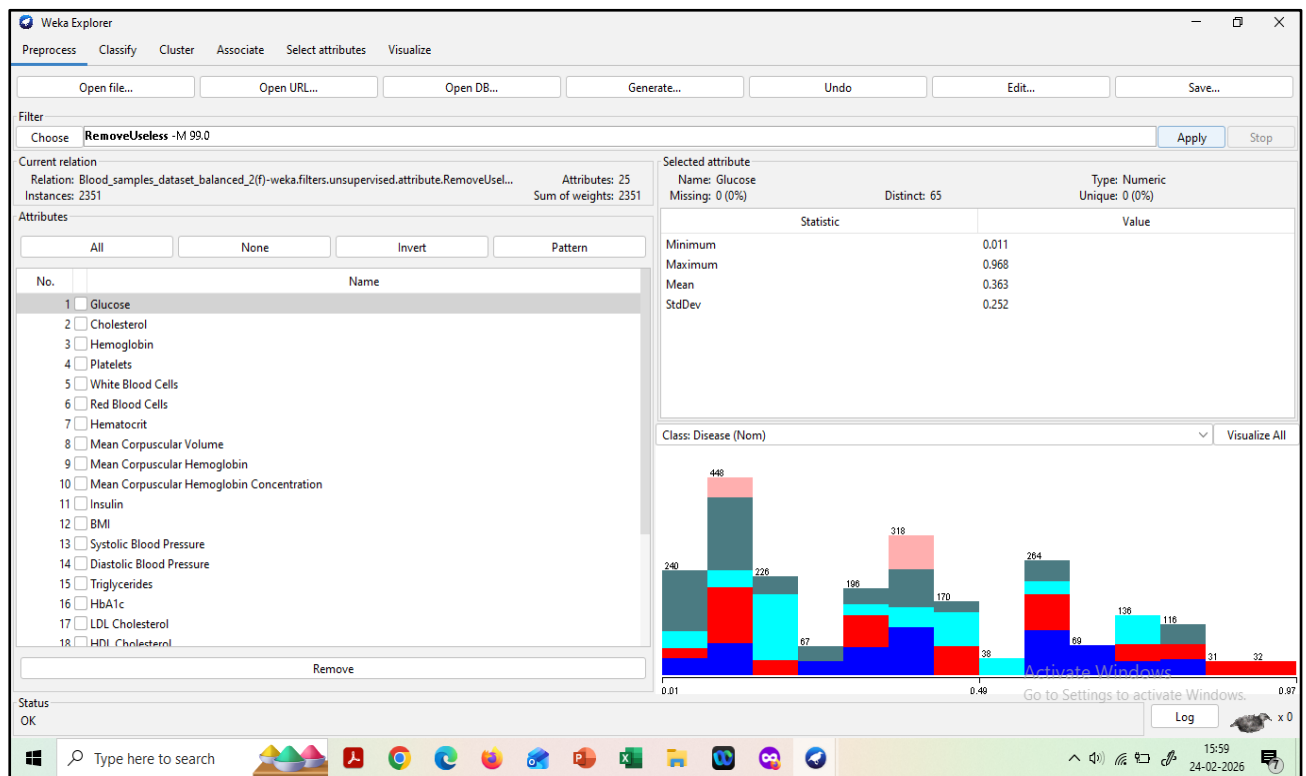
Healthcare datasets are collected from Kaggle repositories containing disease-related blood 25 parameters such as:

- Cholesterol
- Hemoglobin
- Platelets
- White Blood Cells
- Insulin

Step 2: Data Preprocessing

Data preprocessing is performed using WEKA software:

- Data cleaning
- Missing value handling
- Normalization
- Noise removal



Step 3: Machine Learning Model Development

The following machine learning models are trained and tested:

- Naive Bayes
- J48 Decision Tree
- Random Forest

Step 4: Prediction & Evaluation

The models are evaluated based on:

- Accuracy
- Recall
- Precision
- Disease prediction performance

5. Methodology

The proposed methodology follows a supervised machine learning approach.

Dataset Collection



Data Preprocessing



Feature Selection



Model Training



Disease Prediction



Performance Evaluation

The Random Forest model achieved the highest predictive performance compared to other algorithms.

6. Results and Discussion

The machine learning models were compared based on prediction accuracy and recall.

Model	Accuracy	Recall	Performance
Naive Bayes	87%	Medium	Moderate
J48	87.02%	Medium-High	Good
Random Forest	100%	High	Best

The Random Forest algorithm provided the highest accuracy due to its ensemble learning capability and better handling of healthcare datasets.

7. Applications of Predictive Healthcare Analytics

Predictive healthcare systems can be used in:

- Early disease diagnosis
- Hospital management systems
- Patient monitoring systems
- Medical decision support systems
- Personalized healthcare planning

8. Advantages of Proposed System

- Faster disease prediction
 - Improved healthcare decision-making
 - Reduced treatment costs
 - Early risk identification
 - High predictive accuracy
-

9. Future Scope

Future improvements can include:

- Integration with real-time hospital databases
 - Deep learning-based disease prediction
 - IoT healthcare monitoring systems
 - Mobile healthcare applications
 - Cloud-based healthcare analytics platforms
-

10. Conclusion

Predictive Healthcare Analytics plays an important role in improving modern healthcare systems through machine learning technologies. This survey paper discussed existing healthcare prediction methods including Naive Bayes, J48, and Random Forest algorithms. The proposed system uses blood parameter datasets and machine learning models to predict diseases effectively. Experimental results showed that Random Forest achieved the best prediction accuracy. Therefore, predictive analytics can significantly improve early disease detection, reduce healthcare costs, and support evidence-based medical decisions.

References :

1. Han, J., Kamber, M., & Pei, J. *Data Mining Concepts and Techniques*.
2. Tom Mitchell, *Machine Learning*, McGraw-Hill.
3. WEKA Data Mining Software Documentation.
4. Kaggle Healthcare Datasets.
5. Research papers on Machine Learning in Healthcare Analytics.